The Fine-Structure Constant as a Size Ratio

Atomic and Nuclear Binding in One Coulomb Picture

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Mathematical theory and formulation by the author; manuscript developed in collaboration with Claude (Anthropic). See Section 10.

Abstract

We observe that a single Coulomb / multiphase-continuum model (RealQM) describes *both* atomic and nuclear binding through a **variable-size electron**, using only three inputs — the charge / Coulomb potential e^2, an electron kinetic coefficient κ, and the proton's finite size R_p — with *no mass and no strong force.* The electron has no fixed size: it adapts to its environment, spreading to the **atomic** scale $a_0 = 2\kappa/e^2$ when bound to a single proton's charge, and compacting to the **nuclear** scale, set by the proton-size cage, where a flat caged density on a multiphase free boundary carries *zero kinetic energy.* The nuclear/atomic size ratio is then the fine-structure constant squared,

$$\alpha^2 = \frac{R_p}{a_0} = \frac{\text{proton size}}{\text{atomic size}},$$

so α is reinterpreted **geometrically** — the proton-to-atom size ratio, not a coupling constant, and requiring no c. The measured deuteron/hydrogen size ratio matches $1/\alpha^2$ to within $\sim$30% (the deuteron sitting at the classical electron radius). Mass appears nowhere in this electrostatic structure; it is a separate, inertial/gravitational quantity (the matter-wave's inverse size). Finally, since the proton is rigid (fixed size) while the electron is fluid (variable size), α measures the electron's *dynamic range* across scales — and the smallness of α is what permits the nested, multi-scale structure of matter.

Status. A structural/interpretive proposal within the RealQM program. The claim is the *relation* $\alpha^2 = R_p/a_0$ and its consequences; the underlying primitive (the proton's finite size, or equivalently the classical radius) and the numerical realisation are stated as open (Section 8). A speculative cosmological extension is noted only briefly (Section 9).

Keywords: fine-structure constant; atomic and nuclear scales; Coulomb binding; multiphase continuum quantum mechanics; classical electron radius; proton radius; RealQM.

1 Introduction

Why is the nucleus roughly 10^4–10^5 times smaller than the atom that contains it? In standard physics the atomic scale is electromagnetic (set by $\hbar, m, e$) and the nuclear scale is set by the strong force — two unrelated sectors — and the fine-structure constant $\alpha = e^2/\hbar c$ enters as the coupling of the electromagnetic interaction. Here we point out that within the RealQM formulation of quantum mechanics as multiphase Coulomb continuum mechanics [1, 2] — unit-charge densities on non-overlapping domains separated by Bernoulli free boundaries, minimising the Coulomb energy — both the atomic and the nuclear scale arise from *one* mechanism with a *variable-size* electron, and α acquires a direct geometric meaning: the ratio of the proton size to the atomic size. The argument is elementary and uses no relativity; we keep its open points explicit.

2 The model and its three inputs

An electron is a normalised charge cloud ψ ($\int \psi^2\, dV = 1$) of characteristic size L, with energy

$$E = \underbrace{\kappa \int |\nabla\psi|^2\, dV}_{\text{kinetic}} \; - \; \underbrace{Z\,\frac{e^2}{L}}_{\text{Coulomb}}\,, \qquad \kappa = \frac{\hbar^2}{2m}.$$

The whole electrostatic structure of atom and nucleus needs just three inputs: **(i)** the charge / Coulomb potential e^2, **(ii)** the kinetic coefficient κ, and **(iii)** the proton's finite size R_p. *Mass does not appear* (it is replaced by the kinetic coefficient κ); neither does the strong force, nor c. Mass and gravity form a separate sector, discussed in Section 6.

3 The variable-size electron: two scales

The same electron takes two forms, depending on how it is bound.

Outside (atomic). Bound to a single proton, the electron must *taper to zero* in the surrounding vacuum. A varying profile carries kinetic energy $\sim\kappa/L^2$, and minimising $\kappa/L^2 - e^2/L$ gives

$$L = a_0 = \frac{2\kappa}{e^2} \quad \Big(= \frac{\hbar^2}{me^2}\Big),$$

the atomic (Bohr) scale — set by the proton's *charge*. To the atomic electron the proton's finite size is irrelevant (it is 10^4–10^5 times smaller, effectively a point).

Inside (nuclear). Caged between/among protons, the electron sits on its own domain with a multiphase *free boundary* to the proton domains: its density *does not drop to zero* but hands off to the proton density. A *flat* caged density then has $\nabla\psi = 0$ throughout, hence **zero kinetic energy** — there is no kinetic penalty for being small. (The familiar obstacle "confinement to size L costs κ/L^2" applies to a *vacuum* electron forced to vanish at its edge, not to a caged one whose density stays finite at the free boundary.) With no kinetic pressure to spread it, the electron's size is then set by the *proton's finite size* — the cage floor that stops collapse — and its binding is Coulomb at that size, $\sim e^2/R_p \sim$ MeV at fm.

So the electron is **fluid** (it adapts: a_0 outside, R_p inside) while the proton is **rigid** (fixed size). The crucial factor distinguishing the two scales is *proton charge vs. proton size.*

4 α as a size ratio

Combining the two scales, the nuclear/atomic size ratio is

$$\boxed{\alpha^2 = \frac{R_p}{a_0} = \frac{\text{proton (nuclear) size}}{\text{atomic size}}, \qquad \alpha = \sqrt{R_p/a_0}.} \tag{1}$$

That is, α is **the ratio of the proton size to the atomic size** — a geometric quantity, not a coupling constant, and with *no* c. It coincides with the textbook $\alpha = e^2/\hbar c$ precisely when R_p equals the classical radius $r_e = e^2/mc^2 = \alpha^2 a_0 \approx 2.8$ fm. The electron's three characteristic lengths form the ladder

$$a_0 : \bar{\lambda}_C : r_e \;=\; 1 : \alpha : \alpha^2, \qquad \bar{\lambda}_C = \hbar/mc \text{ (Compton)}, \quad r_e = e^2/mc^2 \text{ (classical)},$$

the three ways to build a length from $(\hbar, m, c, e)$, each a factor α apart. The model thus reframes "why $\alpha \approx 1/137$" as "*why is the proton* $\sim$*fm while the atom is* $\sim$*Å*" — relocating the question to the proton size (an input), and interpreting α itself as the proton-to-atom size ratio.

5 Empirical check

The relation is directly testable against measured sizes:

	size	in fm
H atom	Bohr radius $a_0 = 0.53$ Å	$\approx$ 53,000
deuteron nucleus	rms radius $\approx$2.1 fm	$\approx$ 2.1

The ratio is deuteron/H $\approx 4 \times 10^{-5} \approx 1/25{,}000$, to be compared with $\alpha^2 \approx 5 \times 10^{-5} \approx 1/18{,}800$ — **agreement to** $\sim$30%, with the deuteron sitting essentially at the classical electron radius ($r_e = 2.8$ fm). So the nuclear/atomic size ratio comes out as $1/\alpha^2$ from charge, $\hbar$ and the proton size alone, with no strong force. (The deuteron is unusually loosely bound, so this is the *scale/ratio* being right, not a precise fit.)

6 Mass is a separate quantity

Nothing above used mass: the electrostatic atom and nucleus are built from charge and κ. Mass is **inertia** — the localisation (frequency / inverse size) of the matter wave, the dispersion gap $\omega_0 = mc^2/\hbar$, equivalently $m = \hbar/c\,\bar{\lambda}_C^{-1}$. It is a *separate, gravitational* quantity (signed, hence not the positive-definite electromagnetic field energy), and it couples to the electrostatic sector only through energy, $E = mc^2$. Electromagnetism (charge) is an interaction; mass is the inverse size of each matter wave — which is why "the electromagnetic atom has no room for mass."

7 Complexity from the size asymmetry

The structure-building asymmetry between proton and electron is not one of *charge* (charge is exactly balanced, $\pm e$) but of *size*: the proton is fixed-and-small (a rigid anchor), the electron is variable. The fixed proton supplies anchors; the variable electron *spans scales* — spreading to bind atoms and molecules, compacting to bind nuclei — producing the nested, multi-scale hierarchy that *is* complexity. With a fixed size for both, only one scale exists and nothing complex is built. And α measures exactly this asymmetry: $\alpha^2 = R_p/a_0$ is the electron's *inverse dynamic range*, a factor $1/\alpha^2 \approx 18{,}800$ between its nuclear and atomic phases. **The smallness of α is the engine of complexity**; equal fixed sizes would build nothing.

8 Open problems

1. **The nuclear primitive.** The relation $\alpha^2 = R_p/a_0$ takes the proton's finite size R_p as the input nuclear length. Equivalently one may take the relativistic classical radius $r_e = \alpha^2 a_0$ (Coulomb self-energy = rest energy, in the sense of Many-Minds Relativity [3]). Which primitive is fundamental — proton size or relativistic — is open. Either way the value of α is *not* derived, only reinterpreted as a ratio.

2. **Numerical realisation.** Whether the multiphase solver realises a flat, zero-kinetic-energy caged electron at the proton scale has not been cleanly confirmed; direct relaxation runs were inconclusive (non-convergence at small cage size), and a symmetric proton cage with a stable flat density is the proper test.

3. **Equal vs. unequal sizes/coefficients.** The picture is cleanest with one fixed proton size and a variable electron; reconciling this with the measured proton/electron mass ratio (1836) is left to the broader program.

9 Speculative outlook

The same ingredients extend, far more speculatively, to a cosmological narrative — equal-mass $\pm$ charges read off as the curvature of two fluctuating potentials, a gravitational collapse–bounce with negative-mass regions acting as dark energy, and a primordial helium fraction $Y \approx 0.245$ from a compact/free-electron freeze-out. That extension is developed, with its (substantial) open problems stated up front, in a companion essay [4]; it is mentioned here only to indicate scope and is not part of the present, structural claim.

10 Collaboration

The mathematical theory and the proposal are the author's. The manuscript was developed in dialogue with Claude (Anthropic), which assisted in articulating the argument and surfacing its quantitative checks and open problems. We present the human–AI collaboration transparently as part of the working method.

References

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[3] C. Johnson, *Many-Minds Relativity* — light as a medium wave ($c = \sqrt{T/\mu}$), mass as field self-energy ($P = mc \Rightarrow E = mc^2$), and a critique of Einstein's special relativity.

[4] C. Johnson, *Real Cosmology: a proton–electron universe from two fluctuating potentials* (companion essay), same repository.